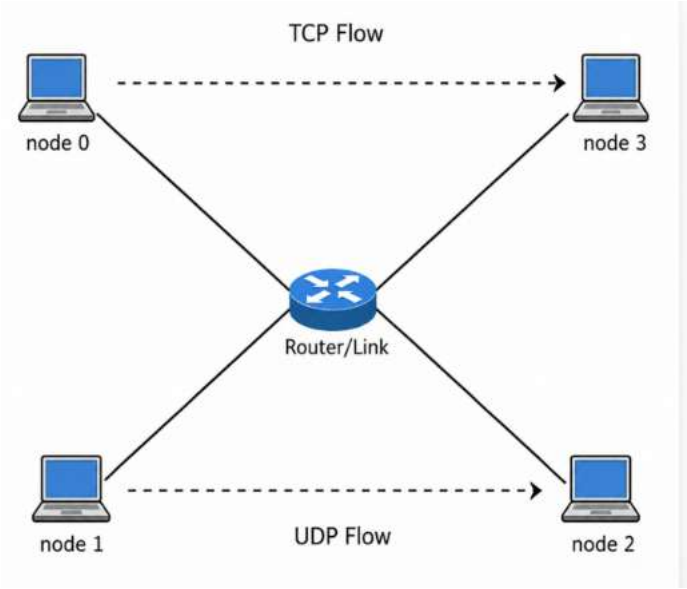
Simulation and Performance Analysis of TCP and UDP Traffic Using NS-3

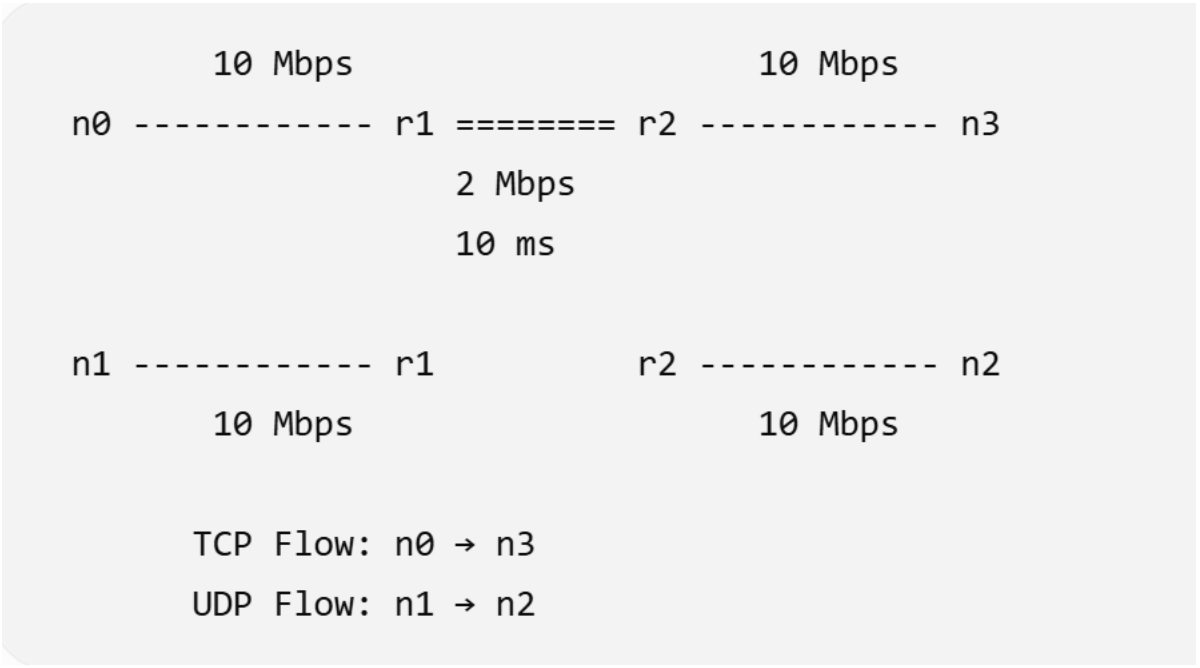
Deadline: 7/9/26 & 9/9/26

Develop an NS-3 program to simulate packet transmission between four end nodes through two intermediate routers as shown below.



- Generate a TCP flow from Node 0 to Node 3 and a UDP flow from Node 1 to Node 2.
- Set the access-link bandwidth to 10 Mbps with 2 ms delay and the bottleneck link bandwidth to 2 Mbps with 10 ms delay.
- Use a DropTail queue of 50 packets
- simulate the network for 20 seconds
- Use a 1024-byte TCP packet and 512-byte UDP packet with a UDP data rate of 1 Mbps.
- Enable NS-3 tracing to record packet-level transmission and reception information.
- Calculate and compare packet loss, average end-to-end delay, throughput, and TCP retransmission rate.
- Repeat the experiment by varying the following parameters

Expected Configuration



Experiment 1 – Bottleneck bandwidth

Run with:

2 Mbps

5 Mbps

10 Mbps

Observe the effect on:

- Packet loss
- Throughput
- Delay
- TCP retransmissions

Extensions (optional) :

Experiment 2 – UDP traffic rate

Run with:

500 Kbps

1 Mbps

2 Mbps

5 Mbps

This is particularly useful because the bottleneck is only **2 Mbps**.

Experiment 3 – Queue size

Run with:

20 packets

50 packets

100 packets

Observe how queue size affects **delay and packet loss**.

Experiment 4 – Propagation delay

Run with:

5 ms

10 ms

20 ms

50 ms

Observe the effect on **end-to-end delay and TCP behavior**.